

Resultados da Aprendizagem

Primitivação imediata

1-Reproduza a seguinte regra:

$$\int f' f^p = \frac{f^{p+1}}{p+1} + C, C \in \mathbb{R}$$

1. $\int x^{10} dx$

Nota: $\int af = a \int f, a \in \mathbb{R}$

2. $\int 3x^{10} dx$

3. $\int (2x)^{10} dx$

4. $\int (x+1)^2 dx$

5. $\int (2x+1)^3 dx$

6. $\int (1+x)^{-2} dx$

7. $\int 3(2x+1)^4 dx$

Nota: $\sqrt[q]{f^p} = f^{p/q}$

8. $\int \sqrt{(2x-5)} dx$

9. $\int \sqrt[3]{3x-2} dx$

10. $\int \sqrt[3]{(x-2)^2} dx$

Nota: $\frac{1}{f^p} = f^{-p}$

11. $\int \frac{1}{(x-3)^2} dx$

12. $\int \frac{1}{(2x-1)^{-2}} dx$

13. $\int \frac{1}{\sqrt{x-1}} dx$

14. $\int \frac{1}{\sqrt[3]{1-2x}} dx$

15. $\int \frac{\pi}{\sqrt[3]{(1-2x)^5}} dx$

Nota: $(f^p)' = p f' f^{p-1}$

16. $\int x(x^2-3)^4 dx$

17. $\int x^2(x^3+1)^3 dx$

18. $\int 2x^3(3-x^4)^5 dx$

19. $\int x \sqrt[5]{x^2-2} dx$

20. $\int (x^3)^4 \sqrt{x^4-2} dx$

21. $\int x \sqrt[5]{(x^2-2)^3} dx$

22. $\int x \sqrt[4]{(1-x^2)^{-3}} dx$

23. $\int \frac{2x^2}{(x^3-3)^2} dx$

24. $\int \frac{2x^3}{(3-x^4)^{-3}} dx$

25. $\int \frac{x}{\sqrt[3]{x^2-2}} dx$

26. $\int \frac{2x^2}{\sqrt[5]{2-x^3}} dx$

27. $\int \frac{2x}{\sqrt[5]{(2-x^2)^3}} dx$

Nota: $(e^f)' = f' e^f$

28. $\int e^x (1+e^x)^{-2} dx$

29. $\int \frac{e^x}{(2-e^x)^2} dx$

30. $\int e^{2x} \sqrt{9+e^{2x}} dx$

31. $\int \frac{e^{-x}}{\sqrt{1-2e^{-x}}} dx$

$$32. \int \frac{e^{x+2}}{\sqrt{1-e^x}} dx$$

$$33. \int \frac{e^{x-1}}{\sqrt{(1-e^x)^3}} dx$$

$$34. \int \frac{e^{\sqrt{x}}(1+e^{\sqrt{x}})^{-2}}{\sqrt{x}} dx$$

Nota: $(\ln(f))' = \frac{f'}{f}$

$$35. \int \frac{1}{x} (\ln(x))^2 dx$$

$$36. \int \frac{(\ln(x+1))^3}{x+1} dx$$

$$37. \int \frac{1}{x \ln^4(x)} dx$$

$$38. \int \frac{1}{x \ln^4(x^2)} dx$$

$$39. \int \frac{1}{x \ln^{-2}(\sqrt{x})} dx$$

$$40. \int \frac{x}{\ln^4(2e^{x^2})} dx$$

$$41. \int \frac{1}{x \sqrt{\ln(x^2)+1}} dx$$

$$42. \int \frac{1}{x \sqrt{(\ln(x)+1)^3}} dx$$

Nota: $(\sin(f))' = f' \cos(f)$

$$(\cos(f))' = -f' \sin(f)$$

$$43. \int \cos(x) \sin^2(x) dx$$

$$44. \int \cos(2x) \sin(2x) dx$$

$$45. \int x \cos(x^2) \sin^4(x^2) dx$$

$$46. \int \frac{\cos(x)}{\sin^2(x)} dx$$

$$47. \int \frac{\cos(3x)}{\sin^4(3x)} dx$$

$$48. \int \frac{\sin(x)}{(\cos(x)-2)^2} dx$$

$$49. \int \cos\left(\frac{x}{2}\right)^3 \sqrt{\sin\left(\frac{x}{2}\right)} dx$$

$$50. \int \sin\left(\frac{x}{3}\right)^3 \sqrt{\cos\left(\frac{x}{3}\right)} dx$$

$$51. \int \sin(3x)^3 \sqrt{2+\cos(3x)} dx$$

$$52. \int x \cos(x^2)^5 \sqrt{(\sin(x^2))^3} dx$$

$$53. \int x \sin(x^2) \left(\sqrt{\cos(x^2)+\pi} \right)^3 dx$$

$$54. \int \frac{\cos(3x)}{\sqrt{\sin(3x)}} dx$$

$$55. \int \frac{3\cos(x+1)}{\sqrt{\sin(x+1)}} dx$$

$$56. \int \frac{x \sin(x^2+1)}{\sqrt[3]{(\cos(x^2+1))^2}} dx$$

$$57. \int \frac{\cos(\sqrt{x}) \sin^3(\sqrt{x})}{\sqrt{x}} dx$$

$$58. \int e^x \cos(e^x) \sin^3(e^x) dx$$

$$59. \int \frac{e^{2x} \cos(e^{2x})}{\sin^3(e^{2x})} dx$$

$$60. \int \frac{e^x \sin(e^x)}{(\cos(e^x)+3)^2} dx$$

$$61. \int \frac{1}{x} \cos(\ln x) \sin^3(\ln x) dx$$

$$62. \int \frac{\cos(\ln x)}{x \sin^3(\ln x)} dx$$

$$63. \int \frac{\cos(\ln x)}{x \sqrt{\sin(\ln x)}} dx$$

$$64. \int \frac{\cos(\ln x)}{x \sqrt{\sin^3(\ln x)}} dx$$

Nota: $\sec(f) = \frac{1}{\cos(f)}$

$$\operatorname{cosec}(f) = \frac{1}{\sin(f)}$$

$$65. \int \cos(x) \operatorname{cosec}^2(x) dx$$

$$66. \int \frac{\sin(2x)}{\sec^3(2x)} dx$$

$$67. \int \frac{\sin(2x)}{\sqrt{\sec^3(2x)}} dx$$

$$68. \int \frac{\cos(3x)}{\sqrt[3]{\operatorname{cosec}^2(3x)}} dx$$

$$69. \int 3x \cos(x^2) \sqrt[3]{\operatorname{cosec}^2(x^2)} dx$$

$$70. \int e^{x+1} \sin(e^x) \sqrt{\sec^3(e^x)} dx$$

$$71. \int \frac{3 \cos(\ln x)}{x \operatorname{cosec}^2(\ln x)} dx$$

$$72. \int \frac{\sin(\sqrt{x})}{\sqrt{x \sec^3(\sqrt{x})}} dx$$

Nota: $(\operatorname{tg}(f))' = f' \sec^2(f)$

$$(\operatorname{cotg}(f))' = -f' \operatorname{cosec}^2(f)$$

$$73. \int \operatorname{cosec}^2\left(\frac{x}{2}\right) \operatorname{cotg}^4\left(\frac{x}{2}\right) dx$$

$$74. \int \sec^2(3x) \operatorname{tg}^3(3x) dx$$

$$75. \int \sec^2(2x) (\operatorname{tg}(2x) + 1)^2 dx$$

$$76. \int \frac{\sec^2(x)}{\operatorname{tg}^3(x)} dx$$

$$77. \int \frac{\operatorname{tg}^3(2x)}{\cos^2(2x)} dx$$

$$78. \int \frac{\pi}{\cos^2(2x) \operatorname{tg}^3(2x)} dx$$

$$79. \int \frac{e^{x+2}}{\sin^2(e^x) \operatorname{cotg}^3(e^x)} dx$$

$$80. \int \frac{\sec^2(\ln(x))}{x \operatorname{cotg}^4(\ln(x))} dx$$

$$81. \int \frac{\operatorname{cosec}^2(\sqrt{x})}{\sqrt{x} \operatorname{tg}^5(\sqrt{x})} dx$$

Nota: $(\operatorname{arctg}(f))' = \frac{f'}{1+f^2}$

$$(\operatorname{arcsen}(f))' = \frac{f'}{\sqrt{1-f^2}}$$

$$82. \int \frac{2}{1+x^2} (\operatorname{arctg}(x))^3 dx$$

$$83. \int \frac{(\operatorname{arcsen}(2x))^2}{\sqrt{1-4x^2}} dx$$

$$84. \int \frac{2e^x}{1+e^{2x}} \operatorname{arctg}(e^x) dx$$

$$85. \int \frac{e^x \sqrt[3]{\operatorname{arcsen}(e^x)}}{\sqrt{1-e^{2x}}} dx$$

$$86. \int \frac{x}{(1+x^4) \sqrt{\operatorname{arctg}(x^2)}} dx$$

$$87. \int \frac{x}{\sqrt{1-x^4}} (\operatorname{arcsen}(x^2))^3 dx$$

$$88. \int \frac{\operatorname{arcsen}(\ln(x))}{x \sqrt{1-\ln^2(x)}} dx$$

$$89. \int \frac{4}{\sqrt{\operatorname{arcsen}(x)(1-x^2)}} dx$$

$$90. \int \frac{\sqrt{x}}{(x+x^2) \sqrt[3]{\operatorname{arctg}(\sqrt{x})+1}} dx$$

$$91. \int \frac{3}{x(1+\ln^2(x)) \sqrt[3]{\operatorname{arctg}(\ln(x))+\pi}} dx$$

Nota: exercícios variados

$$92. \int \frac{\sin(2x)}{(\sin^2(x)-2)^2} dx$$

$$93. \int \frac{e^x \sin(e^x)}{\sqrt[3]{(\cos(e^x)+3)^2}} dx$$

$$94. \int \cos x e^{\sin x} (1+e^{\sin x})^2 dx$$

$$95. \int \frac{\cos x e^{2 \sin x}}{\sqrt[3]{1-e^{2 \sin x}}} dx$$

$$96. \int \frac{\operatorname{tg}(x)}{\ln^2(\cos x)} dx$$

$$97. \int 2 \operatorname{cotg}(x) (\ln(\sin(x)) + 1)^{-3} dx$$

$$98. \int \frac{2 \operatorname{cotg}(x)}{\sqrt[3]{1-\ln(\sin(x))}} dx$$

$$99. \int e^{x+1} \operatorname{cosec}^2(e^x) \operatorname{cotg}(e^x) dx$$

Resultados da Aprendizagem

Primitivação imediata

2-Reproduza as seguintes regras:

$$\int f' e^f = e^f + C, C \in \mathbb{R}$$

$$\int f' a^f = \frac{a^f}{\ln a} + C, C \in \mathbb{R}$$

1. $\int e^{2x-3} dx$

2. $\int x e^{x^2} dx$

3. $\int x^2 4^{x^3+2} dx$

4. $\int \frac{x^2}{e^{x^3}} dx$

5. $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$

6. $\int \frac{10^{\frac{1}{x}}}{x^2} dx$

7. $\int \sin(x) e^{2\cos(x)} dx$

8. $\int \frac{\sin(x)}{e^{\cos(x)}} dx$

9. $\int \sec^2(2x) e^{tg(2x)} dx$

10. $\int \frac{e^{\cotg(2x)}}{\sin^2(2x)} dx$

11. $\int \frac{2^{\arcsen(x)}}{\sqrt{1-x^2}} dx$

12. $\int \frac{e^{\arctg(x)}}{1+x^2} dx$

13. $\int \frac{e^{\operatorname{arccotg}(2x)}}{1+4x^2} dx$

14. $\int \frac{e^{\operatorname{arctg}(\ln(x))}}{x(1+(\ln x)^2)} dx$

$$\int \frac{f'}{f} = \ln|f| + C, C \in \mathbb{R}$$

1. $\int \frac{2}{1-x} dx$

2. $\int \frac{2x^3+x}{x^4+x^2} dx$

3. $\int \frac{e^x}{e^x+1} dx$

4. $\int \frac{e^{x+1}}{e^x+1} dx$

5. $\int \frac{3}{x \ln(x)} dx$

6. $\int \frac{3}{x \ln(\sqrt{x})} dx$

7. $\int \frac{\cos(3x)}{\sin(3x)} dx$

8. $\int \frac{\sin(x)}{\cos(x)+\pi} dx$

9. $\int \frac{\sin(2x)}{\sin^2(x)+4} dx$

10. $\int \frac{e^x \cos(e^x)}{\sin(e^x)} dx$

11. $\int \frac{\sec^2(2x)}{tg(2x)} dx$

12. $\int \frac{\pi}{\sin^2(x) \cotg(x)} dx$

13. $\int \operatorname{cosec}^2(x) tg(x) dx$

14. $\int \frac{\cotg(2x)}{\cos^2(2x)} dx$

15. $\int \frac{tg(\sqrt{x})}{\sqrt{x} \sin^2(\sqrt{x})} dx$

$$16. \int \frac{3}{\sqrt{1-x^2} \arcsen(x)} dx$$

$$17. \int \frac{\pi \cos(x)}{(\sqrt{1-\sin^2(x)}) \arccos(\sin(x))} dx$$

$$18. \int \frac{2x}{(1+x^4) \arctg(x^2)} dx$$

$$19. \int \frac{2e^x}{(1+e^{2x}) \arctg(e^x)} dx$$

$$20. \int \frac{3}{x(1+\ln^2(x)) \arctg(\ln(x))} dx$$

$$\int f' \sen f = -\cos f + C, C \in \mathbb{R}$$

$$\int f' \cos f = \sen f + C, C \in \mathbb{R}$$

$$1. \int \sen(2x) dx$$

$$2. \int \cos(3x+1) dx$$

$$3. \int x \sen(x^2) dx$$

$$4. \int \frac{\sen(\sqrt{x})}{\sqrt{x}} dx$$

$$5. \int \frac{x \sen(\sqrt{x^2+1})}{\sqrt{x^2+1}} dx$$

$$6. \int e^x \cos(e^x) dx$$

$$7. \int \frac{\sen(\ln(x))}{2x} dx$$

$$8. \int \frac{\sen(\ln(x^2))}{x} dx$$

$$9. \int \sqrt{e^x} \sen(\sqrt{e^x}) dx$$

$$10. \int \sen(x) \sen(\cos(x)) dx$$

$$11. \int \sen(2x) \cos(\cos^2(x)) dx$$

$$12. \int \frac{\sen(x)}{\sqrt{\cos(x)}} \sen(\sqrt{\cos(x)}) dx$$

$$13. \int \tg(2x) \sen(\ln(\cos(2x))) dx$$

$$14. \int \frac{4}{\sqrt{1-4x^2}} \sen(\arccos(2x)) dx$$

$$15. \int \frac{e^{x+1}}{\sqrt{1-e^{2x}}} \cos(\arcsen(e^x)) dx$$

$$16. \int \frac{\cos(x)}{2-\cos^2(x)} \cos(\arctg(\sen(x))) dx$$

$$\int f' \sec^2 f = \tg f + C, C \in \mathbb{R}$$

$$\int f' \csc^2 f = -\cotg f + C, C \in \mathbb{R}$$

$$1. \int \sec^2(x+1) dx$$

$$2. \int x^2 \csc^2(x^3) dx$$

$$3. \int \frac{\csc^2(\sqrt[3]{x^2})}{\sqrt[3]{x}} dx$$

$$4. \int \frac{\csc^2(e^{x+2})}{e^{-x}} dx$$

$$5. \int \frac{\sec^2(\ln(x))}{x} dx$$

$$6. \int \frac{\csc^2(\sen(3x))}{\sec(3x)} dx$$

$$7. \int \frac{e^x \csc^2(\cos(e^x))}{\csc(e^x)} dx$$

$$8. \int \frac{1}{4+x^2} \sec^2(\arctg(\frac{x}{2})) dx$$

$$\int f' \csc f = \ln|\csc f - \cotg f| + C, C \in \mathbb{R}$$

$$\int f' \sec f = \ln|\sec f + \tg f| + C, C \in \mathbb{R}$$

$$1. \int \csc(3x+2) dx$$

$$2. \int 3x \csc(x^2+1) dx$$

$$3. \int \frac{\sec(\sqrt{x+2})}{3\sqrt{x+2}} dx$$

$$4. \int e^x \sec(e^{x+1}) dx$$

$$5. \int \frac{\csc(\ln(\sqrt{x}))}{x} dx$$

$$6. \int \cos(2x) \sec(\sen(2x)) dx$$

$$7. \int e^{2x} \operatorname{cosec}(e^{2x+1}) dx$$

$$8. \int \cotg(2x) \operatorname{cosec}(\ln(\operatorname{sen}(2x))) dx$$

$$\int \frac{f'}{\sqrt{1-f^2}} = \operatorname{arcsen}(f) + C, C \in \mathbb{R}$$

$$\int \frac{f'}{1+f^2} = \operatorname{artg}(f) + C, C \in \mathbb{R}$$

$$1. \int \frac{1}{\sqrt{1-(2x)^2}} dx$$

$$2. \int \frac{1}{\sqrt{1-4x^2}} dx$$

$$3. \int \frac{1}{\sqrt{4-4x^2}} dx$$

$$4. \int \frac{x^2}{\sqrt{9-x^6}} dx$$

$$5. \int \frac{x^3}{\sqrt{4-9x^8}} dx$$

$$6. \int \frac{e^x}{\sqrt{1-9e^{2x}}} dx$$

$$7. \int \frac{\operatorname{sen}(x)}{\sqrt{1-\cos^2(x)}} dx$$

$$8. \int \frac{3}{\sec(2x) \sqrt{1-\operatorname{sen}^2(2x)}} dx$$

$$9. \int \frac{4}{x \sqrt{1-\ln^2(x)}} dx$$

$$10. \int \frac{4 \cotg(x)}{\sqrt{1-\ln^2(\operatorname{sen}(x))}} dx$$

$$11. \int \frac{4 \sec^2(2x)}{\sqrt{1-4 \operatorname{tg}^2(2x)}} dx$$

$$12. \int \frac{4}{\sqrt{x} \sqrt{1-4x}} dx$$

$$13. \int \frac{1}{1+4x^2} dx$$

$$14. \int \frac{1}{4+4x^2} dx$$

$$15. \int \frac{x}{9+4x^4} dx$$

$$16. \int \frac{x^2}{4+9x^6} dx$$

$$17. \int \frac{e^x}{1+4e^{2x}} dx$$

$$18. \int \frac{\operatorname{sen}(x)}{1+\cos^2(x)} dx$$

$$19. \int \frac{4}{x(1+\ln^2(x))} dx$$

$$20. \int \frac{4 \cotg(2x)}{(1+\ln^2(\operatorname{sen}(2x)))} dx$$

$$21. \int \frac{4}{x(1+\ln^2(x^2))} dx$$

$$22. \int \frac{x \cos(x^2)}{1+\operatorname{sen}^2(x^2)} dx$$

$$23. \int \frac{\operatorname{sen}(\ln x)}{x(1+\cos^2(\ln x))} dx$$

$$24. \int \frac{4}{\sec(x)(1+\operatorname{sen}^2(x))} dx$$

$$25. \int \frac{3 \operatorname{tg}(2x)}{1+\ln^2(\cos(2x))} dx$$

$$26. \int \frac{6}{\sqrt{x}(1+x)} dx$$

$$27. \int \frac{e^x \operatorname{cosec}^2(e^x)}{1+9 \operatorname{tg}^2(e^x)} dx$$

3- Complete justificando, o espaço assinalado com [] por forma a que possam ser aplicadas as regras da primitivação imediata (Tabelas de Matemática, página 3) indicadas: I. $\int [] \sin(x) dx$ a.R1 b.R2 c.R5 d.R7 e.R18 f.R19	II. $\int [] e^{2x} dx$ a.R1 b.R2 c.R3 d.R5 e.R18 f.R19 III. $\int [] \sec(x) \tan(x) dx$ a.R1 b.R2 c.R5 d.R10 e.R12
4-Identifique em cada uma das seguintes primitivas a(s) expressão(ões) em falta marcadas com [] por forma a que possam ser aplicadas as regras indicadas: $\int \frac{[]}{(1+e^{2x})^2} dx$ (Regra 2) $\int [] \sqrt[3]{\ln^2(x)} dx$ (Regra 2) $\int [] \sqrt{1+\cos(3x)} dx$ (Regra 2) $\int \frac{[]}{\sec^3(x)} dx$ (Regra 2) $\int \frac{[]}{\cot g^5(3x^2)} dx$ (Regra 2) $\int \frac{\text{sen}[] \cos^3(\ln(x))}{[]} dx$ (Regra 2) $\int [] e^{2\text{sen}(x)} dx$ (Regra 3) $\int [] e^{\cos(x^2)} dx$ (Regra 3) $\int \frac{[]}{\text{sen}(e^x + 1)} dx$ (Regra 5) $\int \frac{[]}{\cos(\ln(x)) + 1} dx$ (Regra 5) $\int [] \cot g(e^x) dx$ (Regra 5) $\int [] \cos \sqrt{e^{2x}} dx$ (Regra 6) $\int [] \cos \sqrt{x^2 + 1} dx$ (Regra 6) $\int [] \text{sen}(e^{2x}) dx$ (Regra 7) $\int [] \text{sen}(\ln(x)) dx$ (Regra 7) $\int \frac{[]}{\sqrt{1-e^{4x}}} dx$ (Regra 18)	$\int \frac{[]}{\sqrt{4-9x^2}} dx$ (Regra 18) $\int \frac{[]}{9+4x^2} dx$ (Regra 19) $\int \frac{[]}{1+e^{6x}} dx$ (Regra 19) 5-Identifique em cada uma das seguintes primitivas o(s) erro(s) cometido(s) por forma a que possam ser aplicadas as regras indicadas: $\int e^x \cos(x) \sqrt[3]{\text{sen}(e^x)} dx$ (Regra 2) $\int \text{sen}(\sqrt{x}) \sqrt{\cos(\sqrt{x})} dx$ (Regra 2) $\int \frac{\cos(x)}{\ln^{-2}(\text{sen}(2x))} dx$ (Regra 2) $\int \frac{\cos(x)}{\ln(\text{sen}(x))} dx$ (Regra 5) $\int \frac{e^x}{1+e^{2x}} dx$ (Regra 5) $\int \frac{3}{1+\ln(x)} dx$ (Regra 5) $\int \frac{\text{sen}(x)}{\cos(x)} \text{sen}(\cos(x)) dx$ (Regra 7) $\int \frac{x^7}{\sqrt{4-9x^8}} dx$ (Regra 18) $\int \frac{e^{2x}}{\sqrt{9-e^{2x}}} dx$ (Regra 18) $\int \frac{xe^x}{9+e^{2x}} dx$ (Regra 19) $\int \frac{x \cos(x)}{1+\text{sen}^2(x^2)} dx$ (Regra 19)